SPRINT 1

Requirements Analysis & Work Scope

Project Management System (PMS) | Team 11

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Technology decision	Python + Django (confirmed by the team)
Source of requirements	Project Management System Requirements document (FR1–FR13; NFR1–NFR9)
Sprint dates	Confirm/update in Jira before starting the sprint.

1. Purpose of this document

This document records the requirements analysis for Jira task **SCRUM-3 — Analyze Project Requirements**. It summarizes the approved PMS requirements, proposes a realistic Sprint 1 scope based on the current Jira backlog, and identifies acceptance criteria, dependencies, and decisions the team should confirm. It supplements the existing requirements/SRS; it does not replace them or introduce new product functionality.

2. Project overview

The Project Management System is a web-based application intended to help teams plan, track, and manage projects. The approved requirements describe project and task management, Kanban/workflow, estimation, sprint management, dashboards, notifications, collaboration, search/filtering, reporting, and activity logging.

> **Scope status: The Sprint 1 scope below is a proposal based on the Jira work items shown by the team. The team should confirm the selected requirements before treating them as committed sprint scope. Planned work is not represented as completed work.**

3. Functional requirements inventory

The following inventory preserves the identifiers and intent of the source requirements. It is included for analysis and traceability; it does not mean every item is included in Sprint 1.

ID	Requirement	Source requirement summary
FR1	User Authentication	Register, log in/out, reset passwords; role-based access for Admin, Project Manager, and Team Member.
FR2	Project Management	Create, edit, archive/delete projects; add or remove team members.
FR3	Task/Issue Management	Create, edit, delete and assign tasks with title, description, priority, due date and labels.
FR4	Kanban Board	Show task columns and support drag-and-drop status updates.
FR5	Workflow Management	Define custom workflow states and valid transitions for each project.
FR6	Estimation	Assign story points/time estimates and track estimated versus actual effort.
FR7	Sprint/Milestone Management	Create sprints/milestones, add tasks to a sprint backlog, and close sprints.
FR8	Visibility & Dashboard	Show progress, burndown/velocity charts and task distribution.
FR9	Notifications	Notify users about assignments, status changes, comments and approaching deadlines.
FR10	Comments & Collaboration	Allow task comments and file/document attachments.
FR11	Search & Filter	Search projects/tasks and filter by assignee, priority, status, label or sprint.
FR12	Reporting	Generate downloadable progress, velocity and workload reports.
FR13	Activity Log	Maintain an audit trail of task/project changes.

4. Non-functional requirements inventory

ID	Category	Source requirement summary
NFR1	Performance	Page loads and Kanban updates should respond within 2 seconds under normal load.
NFR2	Scalability	Support a growing number of projects and users without major redesign.
NFR3	Security	Hashed passwords, role-based access, HTTPS, and protection against SQL injection, XSS and CSRF.
NFR4	Usability	Intuitive interface, drag-and-drop, responsive on desktop and tablet.
NFR5	Reliability	At least 99% uptime and graceful error handling.
NFR6	Maintainability	Modular MVC/MVT code, documented APIs and unit test coverage.
NFR7	Portability	Support major browsers and Windows, Linux and macOS.
NFR8	Backup & Recovery	Regular automated backups and recovery after failure.
NFR9	Interoperability	Expose a REST API for integration with tools such as Slack, GitHub or email.

5. Proposed Sprint 1 scope and traceability

The current Jira backlog emphasizes foundation work and project creation/management. The most direct functional requirement represented is FR2. Other backlog items are enabling engineering work and should not be described as completed functional requirements on their own.

Jira work item	Requirement link	Expected Sprint 1 output
SCRUM-3 Analyze Project Requirements	FR1–FR13 / NFR1–NFR9 (review)	Requirements inventory, proposed scope, acceptance criteria, dependencies and open questions.
SCRUM-4 Design System Architecture	Supports selected FRs; NFR6 considered	Agreed high-level Django architecture and module responsibilities.
SCRUM-5 Design Database Schema	FR2; selected data entities	Initial schema/model design aligned to the agreed Sprint 1 features.
SCRUM-6 Set Up GitHub Repository	Engineering/process task	Repository, README, .gitignore and agreed collaboration workflow.
SCRUM-7 Implement Project Creation and Management	FR2	Implement the agreed subset of project creation/management, with validation and persistence.
SCRUM-8 Develop Frontend Foundation	Supports selected FRs; NFR4 considered	Initial UI structure for the selected Sprint 1 workflow.
SCRUM-9 Develop Backend Foundation	Supports selected FRs; NFR6 considered	Django project/app structure, configuration and required backend foundation.
SCRUM-10 Develop Unit & Integration Test Cases	Tests for selected FRs; supports NFR6	Automated unit/integration tests covering normal and relevant invalid/boundary cases.
SCRUM-11 Set Up CI/CD Pipeline	Engineering/process task	GitHub Actions automatically installs dependencies and runs tests on repository changes.
SCRUM-12 Project Documentation	Documentation task	Updated README/setup notes and project records reflecting decisions and actual progress.

Proposed acceptance criteria for the selected implementation work

Area	Acceptance criteria (confirm/adapt with team)
Project management (FR2 subset)	An authorized user can create a project with valid required data; the project is persisted and can be viewed. Invalid/missing required data is rejected with understandable feedback. Only agreed project fields and operations are claimed in scope.
Django foundation	A team member can follow the README to install dependencies, configure the local environment, run migrations if applicable, and start the development server.
Unit/integration testing	Tests run using a documented command; they cover the implemented logic and at least relevant valid and invalid/boundary scenarios.
CI/CD	A push or pull request triggers the workflow; dependencies install and available automated tests run; the workflow clearly reports success or failure.
Documentation	The document matches the actual agreed stack, selected scope and implementation status; unknown decisions are marked TBD rather than guessed.

6. Non-functional requirements addressed in Sprint 1

NFRs are system-wide quality goals. Based on the current Jira work items, the following are reasonable candidates to consider during Sprint 1; their inclusion and verification approach should be agreed by the team.

NFR	Sprint 1 relevance	Evidence / verification
NFR4 — Usability	Consider for the frontend foundation.	Review the initial UI for clear labels, navigation and usable layout. Full responsiveness/drag-and-drop is not claimed unless implemented.
NFR6 — Maintainability	Relevant to Django app structure and testing.	Use a modular app structure, keep setup/API decisions documented, and run automated unit tests for implemented logic.
NFR3 — Security	Relevant if authentication or protected project actions are implemented in this sprint.	Verify authorization and safe credential handling for any implemented authentication. Do not claim full security compliance from repository setup alone.

Other NFRs not yet evidenced by the current backlog

NFR1 (2-second response target), NFR2 (scalability), NFR5 (99% uptime), NFR7 (cross-platform/browser support), NFR8 (backup/recovery), and NFR9 (external REST integrations) are not demonstrated by the current task list alone. Do not mark them complete unless the team adds relevant work and verifies them.

7. Work distribution from the current Jira assignment

Jira ID / task	Assigned member	Responsibility / expected contribution
SCRUM-3 — Analyze Project Requirements	Shivanagouda I Patil	Review requirements, propose scope, document acceptance criteria and dependencies.
SCRUM-4 — Design System Architecture	Shibi R Shetty	Develop architecture design for the agreed Django solution.
SCRUM-5 — Design Database Schema	Shibi R Shetty	Design initial data models/schema for the selected scope.
SCRUM-6 — Set Up GitHub Repository	Shivansh Sharma	Set up repository and coordinate team collaboration.
SCRUM-7 — Implement Project Creation and Management	Shibi R Shetty	Implement the agreed FR2 subset.
SCRUM-8 — Develop Frontend Foundation	Sheetal S Bevinakatti	Develop the agreed initial frontend/UI foundation.
SCRUM-9 — Develop Backend Foundation	Shibi R Shetty	Set up backend structure and required APIs.
SCRUM-10 — Develop Unit & Integration Test Cases	Sheetal S Bevinakatti	Coordinate test cases and test coverage alongside implementation.
SCRUM-11 — Set Up CI/CD Pipeline	Shivansh Sharma	Coordinate automated workflow setup with the developer(s).
SCRUM-12 — Project Documentation	Shivanagouda I Patil	Maintain documentation based on team-approved decisions and actual progress.

8. Dependencies and open decisions

• Confirm the final Sprint 1 FR/NFR scope as a team; the current Jira backlog does not explicitly include a separate FR1 authentication implementation task or FR3 task-management implementation task.

• Confirm and update the sprint dates in Jira. The screenshot shows 1 Oct–8 Oct; verify that these dates reflect the actual agreed schedule before starting the sprint.

• Confirm database choice, Django/Python versions, frontend approach (Django templates or separate frontend), test tooling and local setup instructions.

• Agree which project fields and operations are in scope for SCRUM-7; do not assume all FR2 operations (archive/delete, member add/remove) are included unless planned.

• Define the actual completion criteria and evidence for each Jira item; do not mark work complete before implementation/testing is verified.

9. Completion criteria for SCRUM-3

SCRUM-3 can be considered complete when: (1) the requirements inventory has been reviewed; (2) the team has agreed the Sprint 1 scope; (3) acceptance criteria are documented for the selected work; (4) dependencies/open decisions are recorded; and (5) the findings are added to Jira or shared with the team. Until team agreement is recorded, the scope in this document remains a proposal.

Prepared for team verification. This draft is based on the provided PMS requirements and the Jira assignments shown. It intentionally distinguishes source requirements from proposed Sprint 1 scope and does not claim that planned work is already completed.